

Human and machine age estimation on APPA-REAL database

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1 Introduction

This report evaluates human performance in estimating ages from facial images using the APPA-REAL database and compares it to a neural network-based age predictor. Key findings include:

1. Both human and machine estimates tend to: (i) overestimate the ages of individuals between 15 and 25, and (ii) underestimate the ages of those over 30, with the gap widening as age increases. This may be because teenagers and young adults often aim to look older, while those over 30 typically try to appear younger.
2. Females estimate ages more accurately than males, with a mean absolute error (MAE) of 5.1 compared to males' MAE of 5.6.
3. The age of the person estimating someone else's age has no impact on the accuracy of their predictions.
4. The prediction accuracy, measure by MAE, in estimating the real age by different approaches is: i) human crowd 4.6, ii) machine 4.9, iii) single human 6.0. That is, while the machine predictor outperforms an individual human, the collective wisdom of the crowd remains the most accurate predictor of age, though the advantage over the machine is only slight.
5. Human age predictions vary significantly, even for children. For example, a 7.8-year interval is needed for 90% accuracy when estimating the age of 10-year-olds. For 15-year-olds, a 12.4-year interval is required for the same accuracy. As a result, an interval predictor required to have a high coverage would output very wide, and thus not very informative, intervals.

2 Database

The APPA-REAL database, introduced in Agustsson et al. [2017], comprises 7,591 images featuring a single target face. In a subset of these images, multiple faces are present, with the target face positioned as the largest face at the center. Annotations of the age within the database pertain exclusively to the target face.

Each face stored in the database is linked to the real age (i.e. the chronological age) of the individual portrayed, representing the number of years since their birth. Moreover, the database contains multiple age estimates, referred to as "age guesses," provided by human workers tasked with estimating the age of the depicted person. The APPA-REAL database amalgamates data from three distinct sources:

1. The AgeGuess platform (accessible at AgeGuess.org),
2. An Age Recognition Web application similar to AgeGuess.org, and

3. Workers from Amazon Mechanical Turk.

A subset of the database images and their corresponding real ages were contributed by users of `AgeGuess.org`, potentially introducing noise in the real age annotations.

In total, there are 260,656 age guesses recorded in the dataset. For a subset of these guesses, we have information about the age and gender of the worker responsible. The distribution of workers' ages is depicted in Figure 1a. Meanwhile, Figure 1b showcases the non-uniform distribution of guesses across the images. On average, each image receives 34.3 guesses, with a median of 37. Remarkably, one image has garnered as many as 609 guesses. This particular image, with its extraordinary level of engagement, is spotlighted in Figure 2a, along with an annotated visualization illustrating the distribution of guesses juxtaposed with the real age. This image holds significance as the inaugural one for unregistered users to annotate on the `AgeGuess.org` platform.

The database images are divided into three subsets: training (4,113 images), validation (1,500 images), and testing (1,978 images). Figure 3 illustrates the distribution of both the real and apparent ages across these three subsets.

In the subsequent text, the annotation of the database is denoted as a sequence of $n = 260,656$ tuples

$$((a_i, g_i, \pi_i, w_i^A, w_i^G) \in (\mathcal{A} \times \mathcal{G} \times \mathbb{N} \times \mathbb{N} \times \{\text{male, female, NA}\}) \mid i = 1, \dots, n).$$

Here, $a_i \in \mathcal{A} = \{1, \dots, 100\}$ represents the real age, $g_i \in \mathcal{G} = \{0, \dots, 100\}$ stands for the age guess, $\pi_i \in \mathbb{N}$ denotes the image ID, $w_i^A \in \mathbb{N} \cup \{-1\}$ indicates the worker's age (-1 means not defined) and $w_i^G \in \{\text{male, female, NONE}\}$ is the worker's gender. It is noteworthy that the worker's self-declared ages w_i^A and gender w_i^G are prone to inaccuracies, with some ages clearly being erroneous (values above 100).

In addition to the database annotations, we compute an apparent age for each facial image by averaging all age guesses associated with the depicted individual. Specifically, the apparent age $\bar{g}_\pi \in \mathfrak{R}$ for a face in the image with ID π is calculated as:

$$\bar{g}_\pi = \frac{1}{|\mathcal{I}_\pi|} \sum_{i \in \mathcal{I}_\pi} g_i. \quad (1)$$

Here, $\mathcal{I}_\pi = \{i \in \{1, \dots, n\} \mid \pi_i = \pi\}$ represents the indices of annotations for the image π . Unlike the real age, the facial image inherently contains all the requisite information to determine the apparent age.

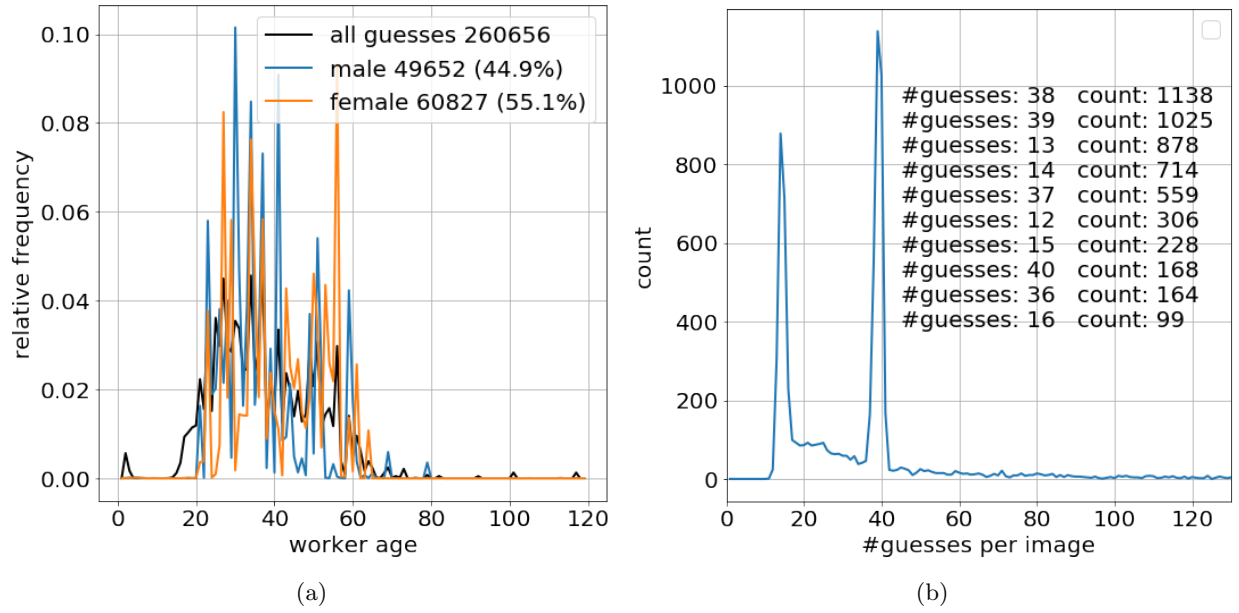


Figure 1: Figure 1a shows the distribution of worker's age computed from all guesses (black) and from subsets of male (blue) and female (orange) guesses where the worker's gender is known. Figure 1b visualizes the distribution of the number of guesses per image, and the top ten most frequent numbers of the guesses.

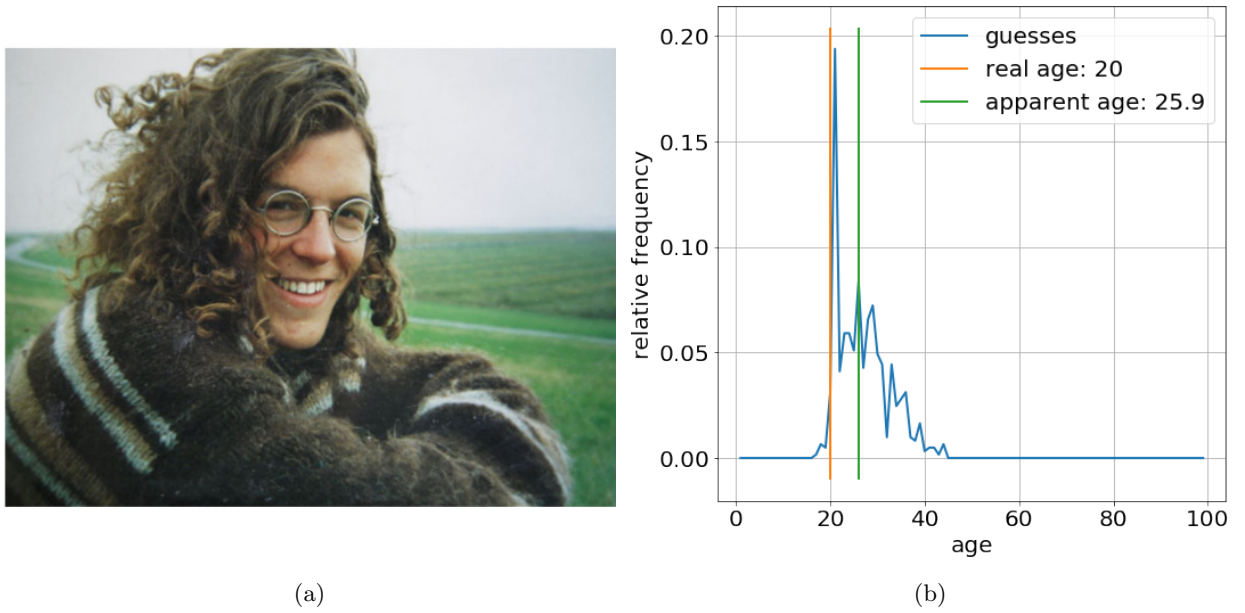


Figure 2: Figure 2a displays database image 005035.jpg, which has the highest number of guesses, totaling 609. Figure 2b provides a visualization of the corresponding distribution of guesses, the real age, and the apparent age, computed as the average over the guesses.

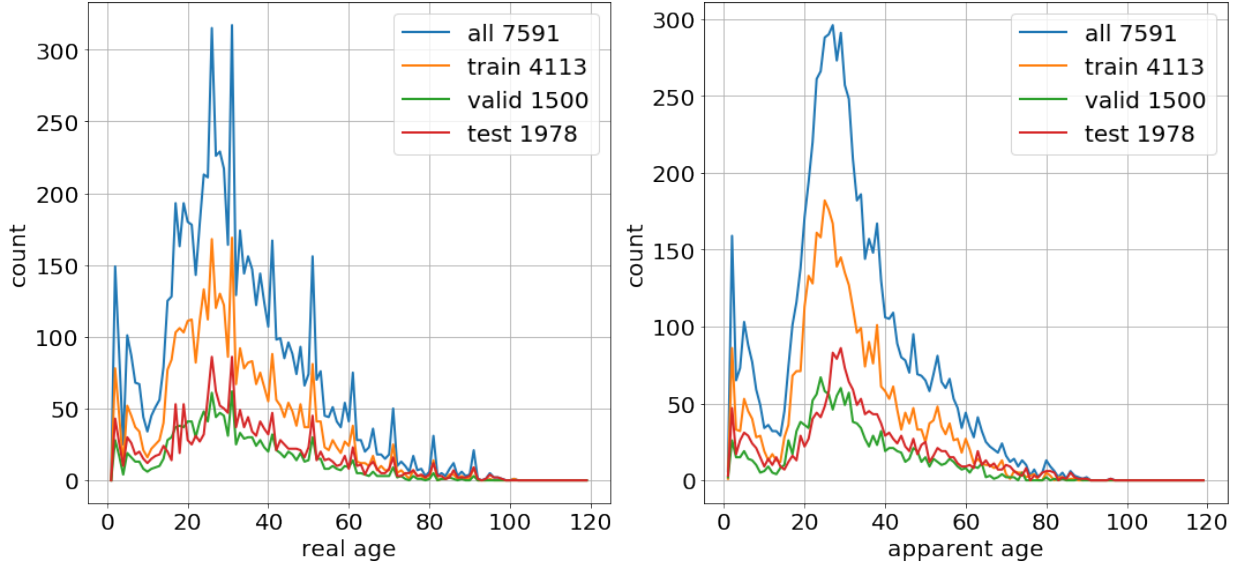


Figure 3: Distribution of the real and the apparent age in the entire image database and training/validation/test subsets.

3 Annotation bias towards ages divisible by 2 and 5

The database contains integer values representing both real ages and age guesses. In Figure 4, the distribution of these ages is depicted. Notably, there is a discernible positive bias toward age labels divisible by 5. This bias is particularly prominent in the case of age guesses, where the frequency of guesses divisible by 5 is 0.27, exceeding the expected frequency of 0.2. The peaks in ages divisible by 5 are also visible in the figure. This observation suggests that individuals, aware of the uncertainty in their predictions, tend to round their estimates to multiples of 5. Interestingly, a similar trend is observed in the distribution of real ages. This discrepancy hints at the presence of annotation noise in the real ages.

Additionally, there appears to be a bias toward even numbers in both real ages and age guesses. The underlying reasons for this bias remain unclear.

	age frequency		
	expected	real	guess
odd	1/2=0.5	0.47	0.47
even	1/2=0.5	0.53	0.53
div by 3	1/3=0.33	0.32	0.32
div by 4	1/4=0.24	0.25	0.26
div by 5	1/5=0.20	0.26	0.27
div by 6	1/6=0.17	0.18	0.17
div by 7	1/7=0.14	0.13	0.14
div by 8	1/8=0.12	0.13	0.13
div by 9	1/9=0.11	0.10	0.11
div by 10	1/10=0.10	0.14	0.13

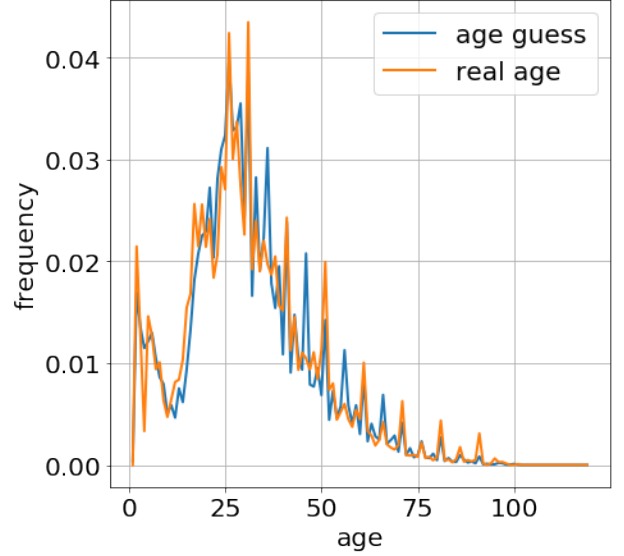


Figure 4: The table and the figure show frequencies of the real ages and the age guesses. Both the real age and age guess are represented by integers. There is an apparent bias towards the ages divisible by 2 and 5.

4 Average apparent age versus real age

In this section, we define the concept of the average apparent age. Specifically, we compute the average apparent age for a given real age a by taking the average of all age guesses associated with database images depicting individuals of that particular real age. That is, given a real age a , the average apparent age is computed as

$$\tilde{g}_a = \frac{1}{|\mathcal{I}_a|} \sum_{i \in \mathcal{I}_a} g_i$$

where $\mathcal{I}_a = \{i \in \{1, \dots, n\} \mid a_i = a\}$ represent the indices of images labeled with the real age a . We compute the average apparent age either from all human guesses or for a subset human workers, e.g. for male/female workers or workers in a given age category.

Each human age guess can be viewed as an independent estimate of the real age. Consequently, we can evaluate the performance of this human prediction by computing the Mean Absolute Error (MAE). The MAE for a subset X of workers is calculated as:

$$\text{MAE}(X) = \frac{1}{|\mathcal{I}_X|} \sum_{i \in \mathcal{I}_X} |g_i - a_i|$$

Here, $\mathcal{I}_X$ denotes the indices of guesses made by workers with attributes X , g_i is the age guess and a_i the real age. For example, to compute the MAE for male workers (MAE(male)) we set $\mathcal{I}_{\text{male}} = \{i \in \{1, \dots, n\} \mid w_i^G = \text{male}\}$.

Figure 5a illustrates the relationship between the average apparent age $\tilde{g}_a$ and the real age, derived from all database guesses. Notably, human age estimations exhibit a tendency to: i) overestimate the age of individuals aged between 15 and 25, and ii) underestimate the age of individuals over 30, with the disparity growing wider as the real age increases. The bias in the real age prediction by humans is better visualized in Figure 7b; see Section 6 for more details. A plausible interpretation of this observation is that teenagers

and young adults aged 15-25 often try to appear older, while individuals over 40 typically strive to maintain a youthful appearance through various means. The human predictor achieves an overall accuracy with the $\text{MAE}(\text{all})$ of 5.3.

In Figure 5b, we compute the average apparent age separately from guesses provided by male and female workers. Remarkably, the correlation between the age guesses and the real age remains consistent for both genders. However, it is worth noting that female workers demonstrate higher prediction accuracy, boasting $\text{MAE}(\text{female})$ of 5.1, compared to male workers, who exhibit a $\text{MAE}(\text{male})$ of 5.6.

Figure 5c illustrates the average apparent age computed from guesses associated with distinct worker age categories. Remarkably, the age estimates across all worker age groups demonstrate striking similarity. Moreover, the prediction accuracy remains stable across various worker age categories, consistently hovering between MAE of 5.2 and 5.3. Our observations contradict the notion that the age of the individual making the age estimates affects prediction accuracy. It is worth mentioning that existing research on this topic presents conflicting viewpoints. Some papers on this topic are listed in Section 5.

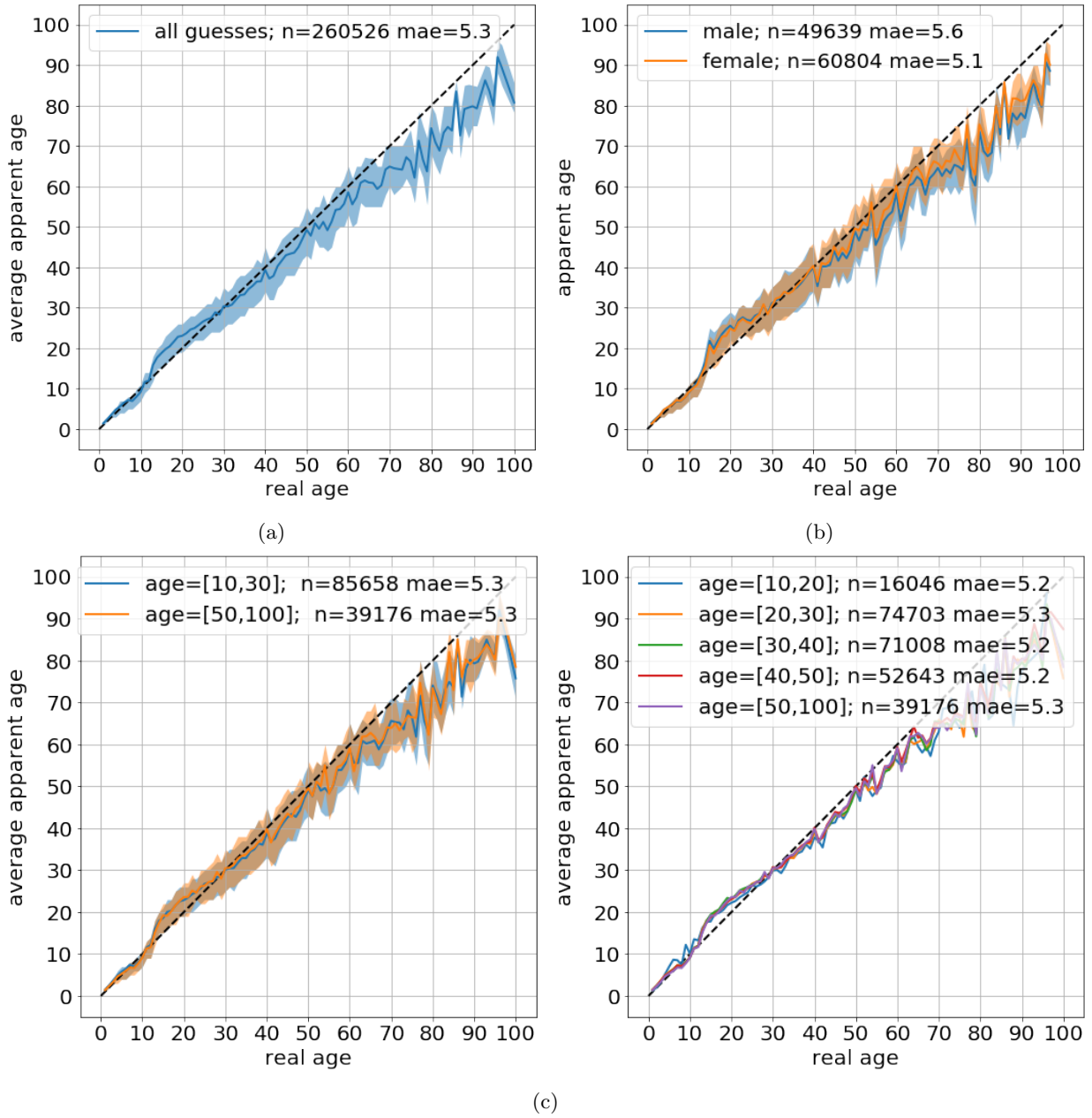


Figure 5: Figure 5a illustrates the relationship between the average apparent age and the real age, computed from all guesses in the database. In Figure 5b, the average apparent age is calculated from guesses where the worker's gender is known. Figure 5c showcases the average apparent age computed from guesses where the worker's age falls within different age categories. The lines correspond to the mean value, while the band is range of values encompassing 50% occurrence.

5 Prior work on human age estimation

Burt and Perrett [1995] This study investigated visual cues to age by using facial composites which blend shape and colour information from multiple faces. To define baseline against which they assess the effect of image manipulations to perceived age, they first measured human age estimates of color images of real faces. These baseline measurements showed that the perceived age of adult male faces was on average closely matched to the real age (over the range 20-62 years). The age rating was performed by 40 subjects. They were split into two groups comprising 20 younger raters (average age 24.5) and 20 older raters (average age 50.8). Each group had equal number of male and female raters. The set of test images (to be rated) consisted of 40 faces with a spread of ages from 20 to 62 years. They fitted a linear function relating the perceived age and the real age. They found the function was the same for the young observers and the older observers. They claim that "There was no tendency for the young observers to overestimate the age of older test faces or vice versa".

Ferguson and Wilkinson [2017] The paper studies the human ability to estimate age of children between 0 and 16 years from their facial images. They use a questionnaire with 30 juvenile face images, where for 20 a single age estimate is required while for 10 faces an age range is asked for (the age groups being 0-5, 6-10, 11-15, 16-20). They collected responses from a total of 504 individuals between age 17 and 82 who, besides answering the age questionnaires, completed a survey. In the survey they were asked about the number and age of children currently living in their household. Their occupation was grouped into three categories: "works with children", "does not work with children" and "child" for individuals under 18 years. The main findings include:

1. The overall accuracy in age prediction is very low. 33% correct responses in age group prediction with only three scoring more than 50% accuracy. In single age question the accuracy 16.5% to 42.8% when allowing for estimates one year above or below the true age.
2. The results suggest the hypothesis that individuals who spend more time in the presence of juveniles will score higher accuracy in the age estimation from juvenile faces is FALSE.
3. *The age and the gender of the participants estimating the age have no effect on the accuracy of the age estimation.*

6 Machine versus human age prediction

In this section, we delve into a comparative analysis of human and machine performance in predicting both the apparent and real age from facial images.

6.1 Predictors

Machine learning models: Our machine predictor employs a multi-layer neural network (MLP) trained on image representations extracted via the FaRL architecture [Paptham and Franc, 2023]. The FaRL representation is initially pre-trained on the 20M LAION database of facial image-text pairs. Subsequently, the MLP undergoes fine-tuning using the training and validation subsets of the APPA-REAL database. We employ two models: i) fine-tuned and evaluated on images annotated with real age and, ii) fine-tuned and evaluated on images annotated with apparent age defined by (1). To assess performance, we compute evaluation metrics from the predicted and ground-truth ages across 1,978 images in the test subset.

Single human predictor: We utilize individual age guesses from the APPA-REAL database as estimates for the real age. It is important to note that the real age remains consistent across all guesses associated with a given database image. This lack of variation is not concerning, as the guesses are expected to be statistically independent. Performance evaluation entails computing the evaluation metrics across 75,464 guesses from the test subset of the database only in order to make the comparison consistent with the machine predictions. This configuration is referred to as the single human predictor and is used as a baseline for the real-age prediction.

Human crowd predictor: In addition to the single human predictor, we employ the apparent age (1) as the predictor of the real age. This baseline is denoted as the human crowd predictor because the apparent age is calculated as the average across all human age guesses.

Real age as the predictor of the apparent age The real age is used as a predictor of the apparent age.

6.2 Evaluation metrics

Let $((a_1, \hat{a}_1), \dots, (a_l, \hat{a}_l))$ be a sequence of l pairs where each pair $(a_i, \hat{a}_i)$ is made up of the real age and the predicted age obtained from a given age predictor. Note that $l = 1,978$ when computing the metrics for the machine predictor and the human crowd predictor, while $l = 75,464$ when evaluating the single human predictor.

We are going to use the following metrics. First, the mean absolute error

$$\text{MAE} = \frac{1}{l} \sum_{i=1}^l |\hat{a}_i - a_i|.$$

The mean absolute error provides an overall summary of the model performance. However, it does not show the performance for different age categories. Therefore, we also use MAE for a given age category

$$\text{MAE}(a) = \frac{1}{n_a} \sum_{i \in \mathcal{I}_a} |\hat{a}_i - a|$$

where $\mathcal{I}_a = \{i \mid a_i = a\}$ denotes indices of test pairs where the real age a_i equal a . In addition, we measure the average prediction bias defined as

$$\text{Bias} = \frac{1}{l} \sum_{i=1}^l (\hat{a}_i - a_i)$$

which measures systematic over/under-estimation of the real age. Finally, we show the bias for a given age category

$$\text{Bias}(a) = \frac{1}{n_a} \sum_{i \in \mathcal{I}_a} (\hat{a}_i - a).$$

Besides the performance on the real age prediction, we use the same metrics to evaluate the prediction of the apparent age in which case the sequence $((a_1, \hat{a}_1), \dots, (a_l, \hat{a}_l))$ is replaced by $((g_1, \hat{g}_1), \dots, (g_l, \hat{g}_l))$, where g_i stands for the true apparent age for i -th image and $\hat{g}_i$ for its estimate. When evaluating the apparent age prediction, the number of examples $l = 1,978$ for both the machine predictor and apparent-by-real-age predictor.

6.3 Results

Figure 6 displays the performance comparison between the machine predictor and the single human predictor in terms of apparent age prediction. Additionally, Figure 7a illustrates the performance of the machine predictor, single human predictor, and human crowd predictor on real age prediction. A summary of the overall performance, including Mean Absolute Error (MAE) and bias, is presented in Table 1. The results reveal that:

Machine vs. single human. It is evident that the machine consistently surpasses the single human predictor in terms of MAE for both real and apparent age prediction. Notably, the machine exhibits significantly superior performance in real age prediction, achieving an MAE of 4.9 compared to 6.0 achieved by the single human.

Machine vs. human crowd. In real age estimation, the human crowd surpasses the machine. The MAE for the human crowd is 4.6, whereas the machine achieves 4.9. While the gap is not substantial, it is unlikely to be solely attributed to the statistical error (randomness in the data).

Estimation bias. The tendency of the human age estimations to i) overestimate the age of individuals aged between 15 and 25, and ii) to underestimate the age of individuals over 30 was discussed above. It can be seen that the bias of the machine estimates follows the very same pattern. This behaviour happens both in the real and the apparent age prediction.

Apparent vs. real age prediction. The estimations of the apparent age exhibit notably superior accuracy when contrasted with real age estimation. This discrepancy is unsurprising, given that apparent age relies solely on facial features. Conversely, facial images often lack the depth of information necessary for precise real age estimation, except in rare instances such as when a birthday cake with candles is present. The machine accuracy is significantly higher than the accuracy achieved when using the real age as the predictor of the apparent age.

Comparison to existing works For comparison, we have included results from Agustsson et al. [2017], who published the APPA-REAL database. Their approach utilizes ensemble CNNs for prediction models. Here, we present only the top-performing results from their paper: the best prediction models achieve a Mean Absolute Error (MAE) of 5.3 for real age prediction and 4.1 for apparent age prediction.

Real age prediction		
	MAE	bias
Machine	4.9	0.40
Single human	6.0	-0.18
Human crowd (apparent age)	4.6	-0.17
Apparent + Residual DEX [Agustsson et al., 2017]	5.3	NA

Apparent age prediction		
	MAE	bias
Machine	3.4	-0.49
Real age	4.6	0.17
Apparent DEX [Agustsson et al., 2017]	4.1	NA

Table 1: The top table shows the performance of the machine, single human and human crowd in predicting the real age. The performance is shown in terms of the Mean Absolute Error (MAE) and the average bias. The bottom table shows the performance of the machine and the real age in predicting the apparent age.

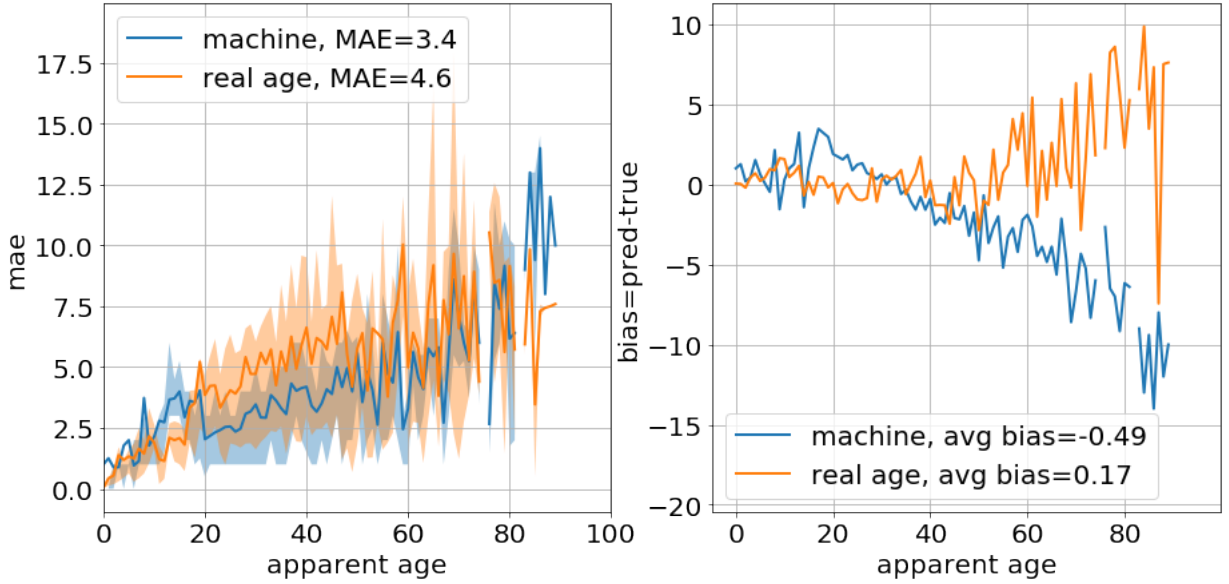


Figure 6: Apparent age prediction: performance of the machine and real age used as a predictor. The left and the right figure shows MAE and the prediction bias conditioned on the real age, respectively.

7 Interval estimate of the apparent age

In this section, we delve into the task of estimating the likely human age guess for a given facial image. To achieve this, we utilize an interval predictor that, given a facial image, returns a range containing a specified portion of human age guesses. Let $\mathcal{G} = \{0, 1, \dots, 100\}$ denote the set of possible human age

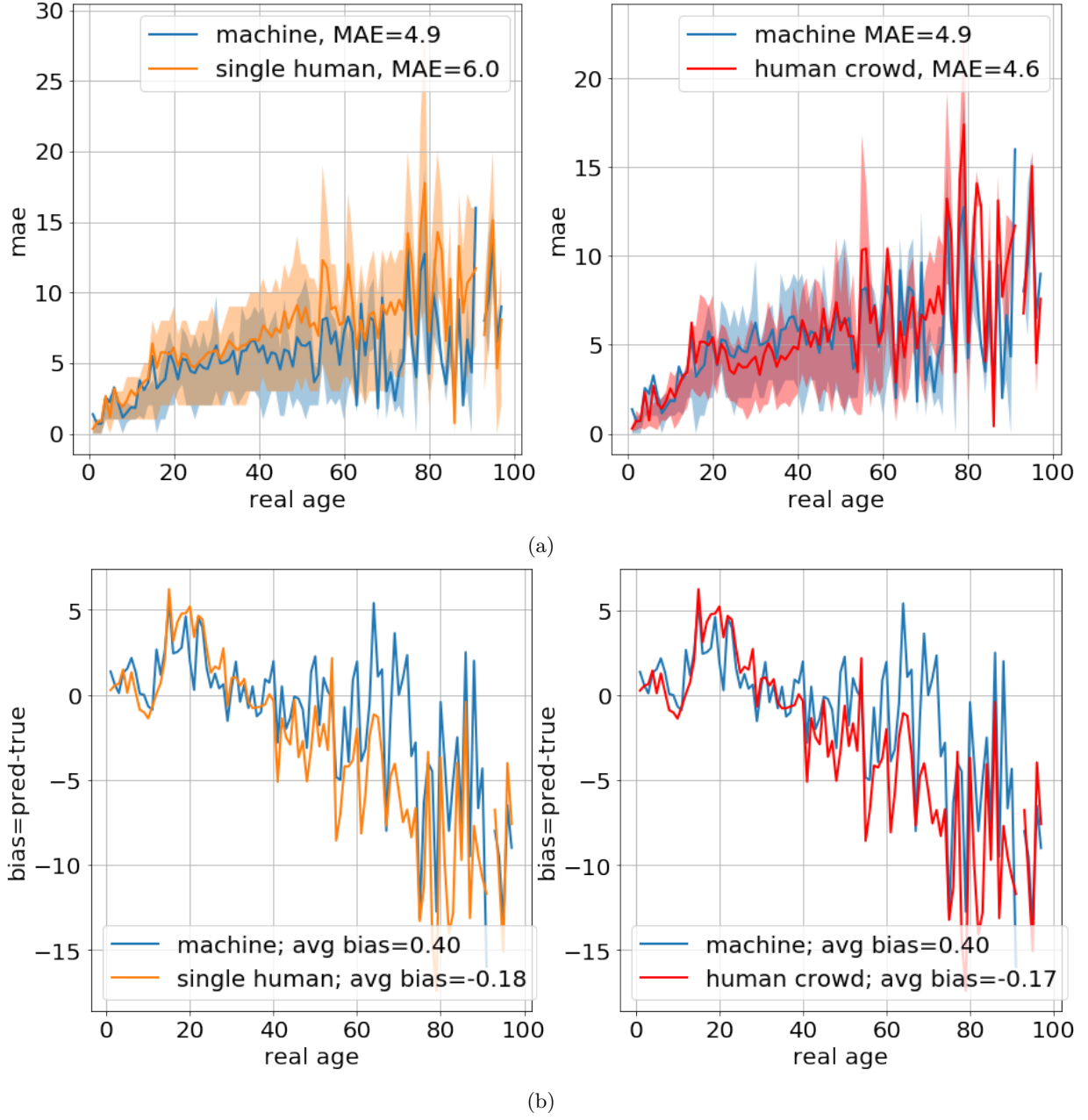


Figure 7: Real age prediction: performance of the machine, a single human and the human crowd. The top figures show MAE as a function of the real age and the bottom figures show the prediction bias.

guesses, $\mathcal{P} = \{(L, R) \in \mathcal{G} \times \mathcal{G} \mid L \leq R\}$ represent the set of all intervals on $\mathcal{G}$, and $\mathcal{X}$ signify the set of all facial images. The interval predictor of the human age guess is a function $C: \mathcal{X} \rightarrow \mathcal{P}$.

7.1 Evaluation metrics for interval predictors

The effectiveness of the interval predictor can be assessed through two key metrics. Firstly, the coverage, which represents the proportion of interval predictions that encompass the real age, is defined as:

$$\rho = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[L_i \leq g_i \leq R_i],$$

where n is the total number of predictions, L_i and R_i denote the lower and upper bounds of the i -th prediction interval, and g_i represents the actual age guess. Secondly, the average interval width of the predictor, denoted as I , is calculated as:

$$I = \frac{1}{n} \sum_{i=1}^n (R_i - L_i + 1).$$

This metric provides insight into the typical range of age predictions made by the interval predictor. Furthermore, we introduce the conditional average prediction interval size, $I(a)$, which is defined as:

$$I(a) = \frac{1}{n_a} \sum_{i \in \mathcal{I}_a} (R_i - L_i + 1).$$

where n_a represents the number of instances where the real age is a , and $\mathcal{I}_a$ denotes the set of indices corresponding to predictions where the real age is a . This metric allows for a more nuanced examination of prediction intervals across different age groups.

7.2 Approximation of an ideal interval predictor

To construct the ideal interval predictor, one requires the true distribution of age guesses. We approximate this ideal predictor using the empirical estimate of the age guess distribution from the APPA-Real database. Evaluating this predictor offers valuable insights into the expected performance when training the interval predictor with data.

Let $p: \mathcal{G} \times \mathcal{X} \rightarrow [0, 1]$ denote the distribution of age guesses made by humans for a given facial image. Consider $\hat{p}(g \mid x)$, where $x \in \{x_1, \dots, x_n\}$, as an empirical approximation of the age guess distribution computed from age estimations in the APPA-REAL database. Utilizing $\hat{p}(g \mid x)$, we can derive an approximation of the perfect interval predictor as follows:

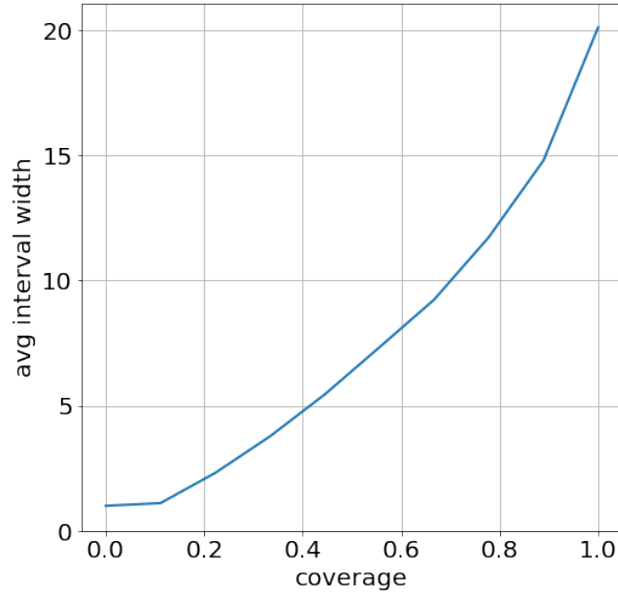
$$C^*(x) = \arg \min_{(L, R) \in \mathcal{P}} (R - L) \quad \text{s.t.} \quad \sum_{g=L}^R \hat{p}(g \mid x) \geq \alpha,$$

where $\alpha \in [0, 1]$ represents a predefined target coverage. For example, for $\alpha = 0.9$ the predicted interval will contain the true age guess in 90% of cases.

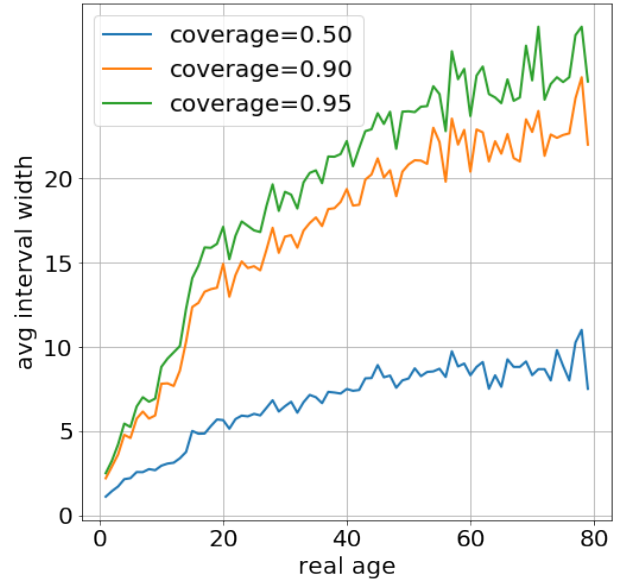
7.3 Results

In Figure 8a, the average interval width is depicted for $C^*(x)$ as the coverage varies across $\alpha \in [0, 1]$. Additionally, Figure 8b presents the conditional average interval width for interval predictors with fixed

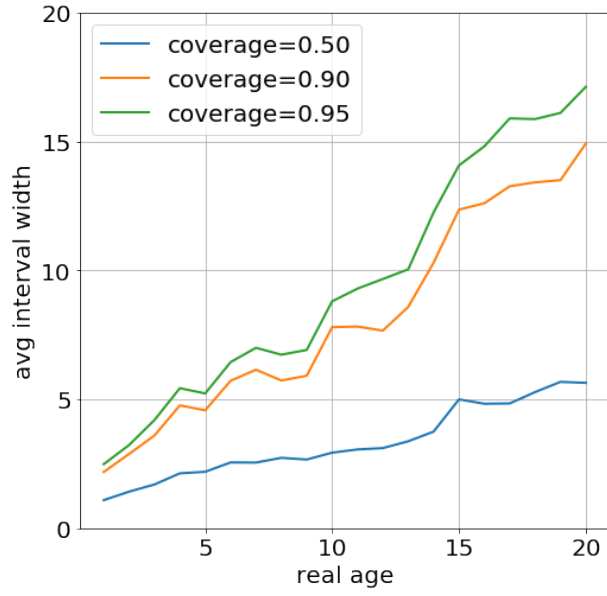
coverages of $\alpha \in \{0.5, 0.90, 0.95\}$. A closer examination of age ranges between 0 and 20 is provided in the zoomed-in version of the same figure, displayed in Figure 8b.



(a)



(b)



(c)

	real age		
	$a = 5$	$a = 10$	$a = 15$
$\alpha = 0.50$	2.2	2.9	5.0
$\alpha = 0.90$	4.6	7.8	12.4
$\alpha = 0.95$	5.2	8.8	14.1

Figure 8: The figures show performance of the approximation of an ideal interval predictor of the human age guess. Figure 8a shows the average interval width as the function of the coverage. Figure 8b and 8c shows the average interval width as a function of the true age for the coverage fixed to $\alpha \in \{0.5, 0.90, 0.95\}$. The table visualizes selected values of the average interval width from Figure 8c

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